'Domain Knowledge

UNSUPERVISED LEARNING

→ We will need large data for better insights

K=? ↳ clusters

TECHNIQUE

- CLUSTERING → K-MEANS, HIERARCHICAL, DBSCAN, GAUSSIAN MIXTURE

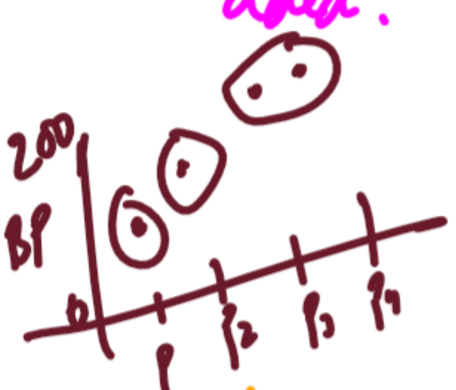
maximize similarity within clusters and minimize similarity b/w diff clusters, revealing hidden patterns in the data.

K-MEANS → divides data into K clusters by minimizing the distance b/w points and their cluster centroids.

HIERARCHICAL → builds a tree of clusters by iteratively merging or splitting data points based on similarity.

DBSCAN → forms clusters from high density regions and identifies sparse points as outliers (noise).

GAUSSIAN MIXTURE → assumes data is generated from a mixture of multiple gaussian distⁿ, each representing a different cluster.



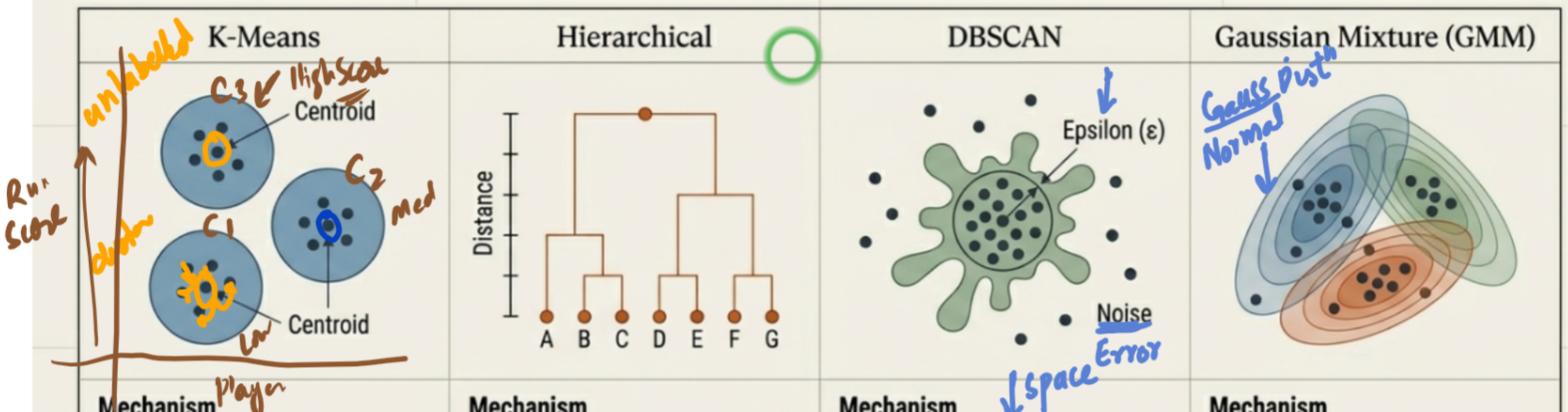
	B.P	Status
P_1	100	N.P
P_2	140	A.P
P_3	170	H.P
P_4	180	S.P

Supervised

	B.P
P_1	100
P_2	140
P_3	170
P_4	180

→ Unsup

The Clustering Algorithm Matrix



	K-Means	Hierarchical	DBSCAN	Gaussian Mixture (GMM)
Mechanism	Minimizes distance to 'k' predefined centroids.	Builds nested tree structures (agglomerative/bottom-up).	Density-Based Spatial Clustering	Probabilistic model assuming overlapping Gaussian distributions.
Strengths	Computationally efficient; excellent for broad global partitions.	High interpretability; no need to predefine the number of clusters.	Excels at finding irregular microclusters and automatically isolating noise/outliers.	Captures complex shapes via soft clustering (probabilistic assignment).
Drawbacks	Forces spherical shapes; highly sensitive to initial parameters.	Computationally expensive for large datasets.	Struggles with varying densities.	Struggles with non-Gaussian datasets.

K-MEAN CLUSTERING

- K-Means aims to minimize the total within-cluster variance:

$$J = \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$

- K : no. of clusters.
- C_i : set of points in cluster i.
- μ_i : centroid (mean) of " ".

INPUT OBSERV

$x - \mu$ Avergy/Mean

$\bar{x}$ → sample ← n

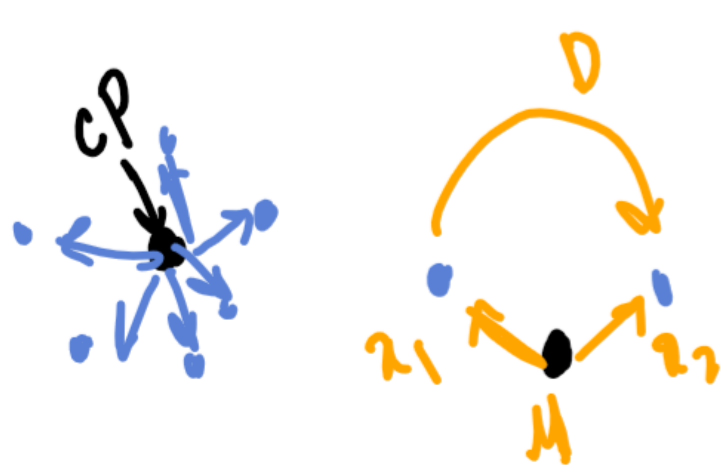
μ → population ← N

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\mu = \frac{\Sigma X}{N}$$

- Distance Measure (Euclidean)

$$d(x, \mu) = \sqrt{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2}$$

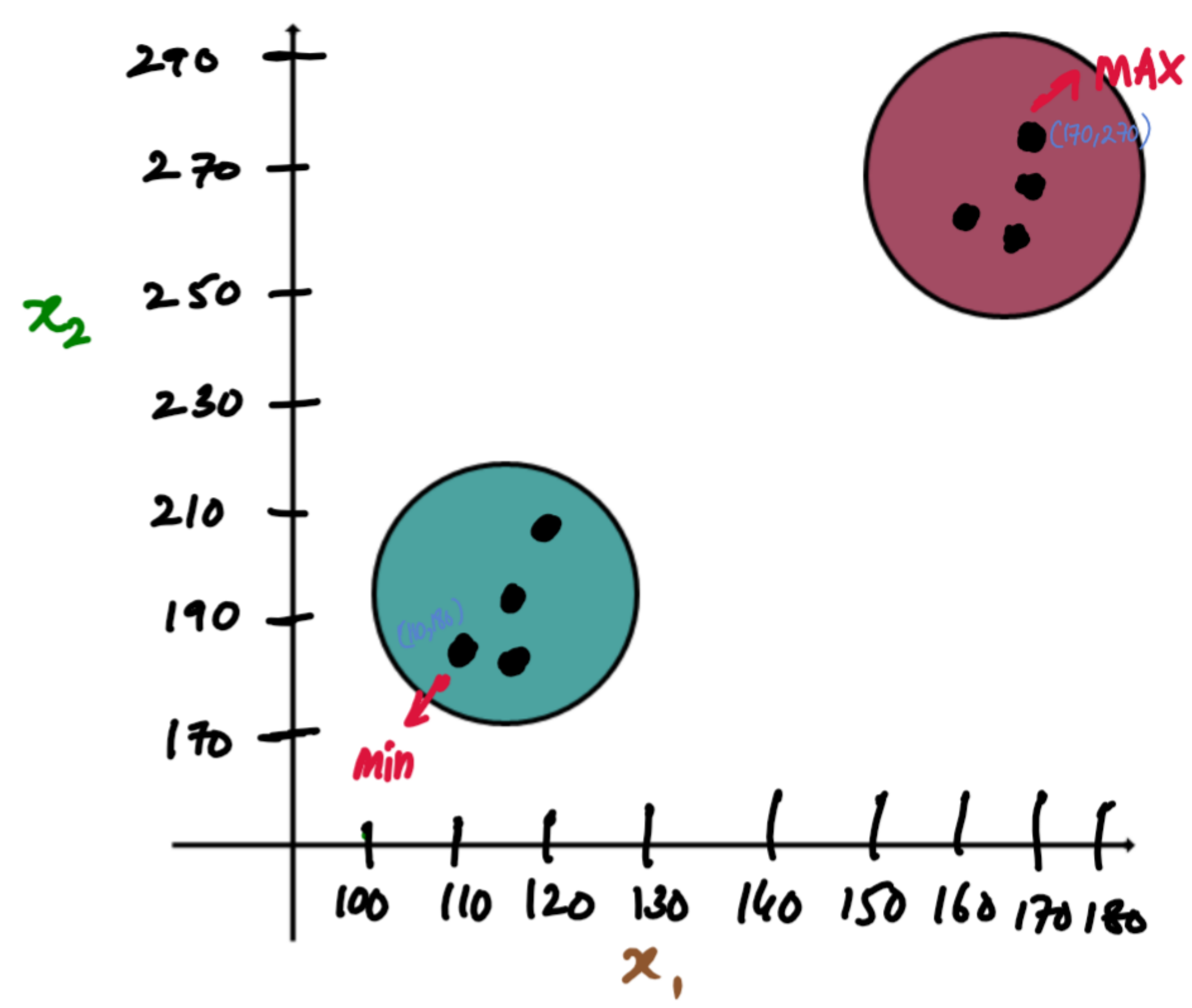


K-M

EXAMPLE:-

A hospital wants to group patients based on health risk factors to priorilize treatment and resourse allocation. They collected dataset as

Patient	x_1: Blood Pres.	x_2: Cholestrol
P_1	110	180
P_2	115	190
P_3	120	200
P_4	160	260
P_5	170	270
P_6	165	255
P_7	118	186
P_8	168	265



STEP-I: Select $K = ?$

$K = 2$: $C_1 \rightarrow$ Low risk, $C_2 \rightarrow$ High risk

STEP-II Calculation of Centroids

$\mu_1 = (\overset{\mu_{1x_1}}{110}, \overset{\mu_{1x_2}}{180}) \longrightarrow C_1$

$\mu_2 = (\overset{\mu_{2x_1}}{170}, \overset{\mu_{2x_2}}{270}) \longrightarrow C_2$

Patient	x_1: Blood Pres.	x_2: Cholestrol	For Centroid μ_1: $d = \sqrt{(x_1-\mu_{1x_1})^2+(x_2-\mu_{1x_2})^2}$	For μ_2: $d = \sqrt{(x_1-\mu_{2x_1})^2+(x_2-\mu_{2x_2})^2}$	Cluster
P_1	110	180	$d = \sqrt{(110-110)^2+(180-180)^2} = \underline{\underline{0}} \downarrow$	$d = \sqrt{(110-170)^2+(180-270)^2} = 90 \uparrow$	C_1
P_2	115	190			
P_3	120	200			
P_4	160	260	$d = \sqrt{(160-110)^2+(260-180)^2} = 94.3$	$d = \sqrt{(160-170)^2+(180-270)^2} = 90.5$	C_2
P_5	170	270			
P_6	165	255			
P_7	118	186			
P_8	168	265			